

Practical Demo: Constructing a Benchmark-Aware Bond Portfolio

From an investable universe to weights, trades, and stress tests

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Goal and Decision

An investment-grade manager has **\$50 million** to invest. The mandate is to earn more carry than a broad bond benchmark without taking an unintended duration, curve, sector, issuer, or

liquidity bet. This demo turns that mandate into an auditable portfolio and an implementable trade list.

The security descriptions and market analytics are a **dated, market-like teaching snapshot**, not executable quotes. In practice, the same table would come from a portfolio system, evaluated pricing service, dealer runs, and reference-data checks.

Workflow stage	In this case
Supplied	Bond terms, yields, spreads, durations, liquidity costs, benchmark and current weights
Constructed	A transparent cell-sampled portfolio and a constrained optimized portfolio
Diagnosed	Factor mismatches, tracking error, concentration, turnover, and transaction cost
Used for	Trade sizing, scenario P&L, and a portfolio recommendation

1 1. Define the Mandate

The optimized portfolio must be fully invested, hold no more than 18% in one security, keep at least 25% in Treasuries, cap corporate exposure at 45%, and stay within 0.05 years of benchmark duration. These are teaching constraints; an actual mandate would also specify ratings, currencies, derivatives, cash, tax, ESG, and counterparty rules.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import minimize

portfolio_value = 50_000_000
position_cap = 0.18
treasury_minimum = 0.25
corporate_cap = 0.45
duration_tolerance = 0.05
```

2 2. Assemble the Investable Universe

Yield and spread are annual percentages expressed as decimals. Duration and spread duration are in years; transaction cost is a one-way estimate in basis points of market value. KRD 2,

KRD 5, and KRD 10 are simplified key-rate duration contributions per dollar invested.

```

universe = pd.DataFrame({
    "Bond": ["UST 2Y", "UST 5Y", "UST 10Y", "Agency 3Y", "Agency 7Y",
            "Bank A 3Y", "Bank B 5Y", "Utility A 7Y", "Industrial A 10Y",
            "Industrial B 5Y", "Telecom A 10Y", "Consumer A 7Y"],
    "Sector": ["Treasury", "Treasury", "Treasury", "Agency", "Agency",
              "Corporate", "Corporate", "Corporate", "Corporate",
              "Corporate", "Corporate", "Corporate"],
    "Cell": ["Gov 1-4", "Gov 5-8", "Gov 9+", "Gov 1-4", "Gov 5-8",
            "Corp 1-4", "Corp 5-8", "Corp 5-8", "Corp 9+",
            "Corp 5-8", "Corp 9+", "Corp 5-8"],
    "Yield": [0.0415, 0.0435, 0.0455, 0.0442, 0.0470, 0.0505,
             0.0528, 0.0555, 0.0585, 0.0540, 0.0600, 0.0562],
    "Spread (bp)": [0, 0, 0, 38, 55, 92, 112, 138, 165, 125, 182, 148],
    "Duration": [1.88, 4.45, 7.85, 2.72, 5.82, 2.65, 4.35, 5.65,
                7.35, 4.18, 7.10, 5.45],
    "Convexity": [4.8, 23.5, 72.0, 9.0, 39.5, 8.5, 22.0, 37.0,
                 62.0, 20.5, 58.0, 35.0],
    "Spread duration": [0, 0, 0, 2.60, 5.65, 2.55, 4.20, 5.45,
                      7.05, 4.02, 6.75, 5.25],
    "KRD 2": [1.70, 0.35, 0.05, 2.30, 0.25, 2.20, 0.45, 0.20, 0.10, 0.50, 0.10, 0.25],
    "KRD 5": [0.18, 3.75, 1.20, 0.42, 4.30, 0.45, 3.35, 3.80, 1.25, 3.10, 1.10, 3.65],
    "KRD 10": [0.00, 0.35, 6.60, 0.00, 1.27, 0.00, 0.55, 1.65, 6.00, 0.58, 5.90, 1.55],
    "Trading cost (bp)": [1, 2, 3, 3, 5, 6, 7, 9, 12, 8, 14, 10],
    "Benchmark weight": [0.10, 0.13, 0.10, 0.09, 0.10, 0.07,
                       0.09, 0.08, 0.07, 0.07, 0.05, 0.05],
    "Current weight": [0.12, 0.10, 0.08, 0.10, 0.08, 0.08,
                     0.10, 0.07, 0.05, 0.08, 0.04, 0.10],
})
assert np.isclose(universe["Benchmark weight"].sum(), 1)
assert np.isclose(universe["Current weight"].sum(), 1)
universe[["Bond", "Sector", "Cell", "Yield", "Spread (bp)", "Duration",
          "Spread duration", "Trading cost (bp)", "Benchmark weight"]]

```

	Bond	Sector	Cell	Yield	Spread (bp)	Duration	Spread duration	Trading cost (bp)
0	UST 2Y	Treasury	Gov 1-4	0.0415	0	1.88	0.00	1
1	UST 5Y	Treasury	Gov 5-8	0.0435	0	4.45	0.00	2
2	UST 10Y	Treasury	Gov 9+	0.0455	0	7.85	0.00	3
3	Agency 3Y	Agency	Gov 1-4	0.0442	38	2.72	2.60	3
4	Agency 7Y	Agency	Gov 5-8	0.0470	55	5.82	5.65	5

	Bond	Sector	Cell	Yield	Spread (bp)	Duration	Spread duration	Trading c
5	Bank A 3Y	Corporate	Corp 1-4	0.0505	92	2.65	2.55	6
6	Bank B 5Y	Corporate	Corp 5-8	0.0528	112	4.35	4.20	7
7	Utility A 7Y	Corporate	Corp 5-8	0.0555	138	5.65	5.45	9
8	Industrial A 10Y	Corporate	Corp 9+	0.0585	165	7.35	7.05	12
9	Industrial B 5Y	Corporate	Corp 5-8	0.0540	125	4.18	4.02	8
10	Telecom A 10Y	Corporate	Corp 9+	0.0600	182	7.10	6.75	14
11	Consumer A 7Y	Corporate	Corp 5-8	0.0562	148	5.45	5.25	10

3 3. Translate Holdings into Portfolio Exposures

For weights w_i and security exposure x_{ik} , portfolio exposure to factor k is $x_{P,k} = \sum_i w_i x_{ik}$. Dollar value of one basis point is $DV01 = V_P D_P(0.0001)$.

```
factor_columns = ["KRD 2", "KRD 5", "KRD 10", "Spread duration"]

def portfolio_summary(weights):
    weights = np.asarray(weights)
    result = {
        "Yield (%)": 100 * weights @ universe["Yield"],
        "Duration": weights @ universe["Duration"],
        "Convexity": weights @ universe["Convexity"],
        "Spread duration": weights @ universe["Spread duration"],
        "Treasury weight (%)": 100 * weights[universe["Sector"].eq("Treasury")].sum(),
        "Corporate weight (%)": 100 * weights[universe["Sector"].eq("Corporate")].sum(),
        "Largest position (%)": 100 * weights.max(),
    }
    result.update({column: weights @ universe[column] for column in factor_columns[:3]})
    result["DV01 ($/bp)"] = portfolio_value * result["Duration"] * 0.0001
    return pd.Series(result)

benchmark_weights = universe["Benchmark weight"].to_numpy()
current_weights = universe["Current weight"].to_numpy()
portfolio_summary(benchmark_weights).round(3)
```

Yield (%)	4.944
Duration	4.842
Convexity	31.455
Spread duration	3.166

Treasury weight (%)	33.000
Corporate weight (%)	48.000
Largest position (%)	13.000
KRD 2	0.722
KRD 5	2.272
KRD 10	1.847
DV01 (\$/bp)	24209.500
dtype:	float64

4 4. Build a Transparent Cell-Sampled Portfolio

Cell sampling first matches broad sector/maturity buckets, then chooses a representative liquid bond in each cell. It is easy to explain, but matching cell weights does not guarantee matching duration or key-rate exposure.

```
cell_weights = universe.groupby("Cell")["Benchmark weight"].sum()
representatives = {
    "Gov 1-4": "UST 2Y", "Gov 5-8": "UST 5Y", "Gov 9+": "UST 10Y",
    "Corp 1-4": "Bank A 3Y", "Corp 5-8": "Bank B 5Y", "Corp 9+": "Industrial A 10Y",
}
sampled_weights = np.zeros(len(universe))
for cell, weight in cell_weights.items():
    bond_name = representatives[cell]
    sampled_weights[universe.index[universe["Bond"].eq(bond_name)][0]] = weight

sampled_table = pd.DataFrame({"Benchmark": benchmark_weights,
                             "Cell sampled": sampled_weights}, index=universe["Bond"])
sampled_table.loc[sampled_table.sum(axis=1).gt(0)].round(3)
```

	Benchmark	Cell sampled
Bond		
UST 2Y	0.10	0.19
UST 5Y	0.13	0.23
UST 10Y	0.10	0.10
Agency 3Y	0.09	0.00
Agency 7Y	0.10	0.00
Bank A 3Y	0.07	0.07
Bank B 5Y	0.09	0.29
Utility A 7Y	0.08	0.00
Industrial A 10Y	0.07	0.12

	Benchmark	Cell sampled
Bond		
Industrial B 5Y	0.07	0.00
Telecom A 10Y	0.05	0.00
Consumer A 7Y	0.05	0.00

5 5. Construct a Constrained Active Portfolio

Let $a = w - w_B$ be active weights and B the factor-exposure matrix. A compact factor-risk approximation is

$$TE^2 \approx (B'a)' \Sigma_F (B'a) + \sum_i a_i^2 \sigma_{\epsilon,i}^2.$$

The objective balances predicted tracking error, turnover, transaction cost, and a modest preference for carry:

$$\min_w TE^2 + \lambda_T \sum_i (w_i - w_i^{current})^2 + \lambda_C \sum_i c_i |w_i - w_i^{current}| - \lambda_Y w' y.$$

This is a teaching risk model. Production construction would use a validated factor covariance matrix, specific-risk forecasts, executable costs, lot sizes, and compliance-tested constraints.

```

exposure_matrix = universe[factor_columns].to_numpy()
factor_vol = np.array([0.0070, 0.0080, 0.0090, 0.0060])
factor_correlation = np.array([
    [1.00, 0.82, 0.62, 0.05], [0.82, 1.00, 0.85, 0.08],
    [0.62, 0.85, 1.00, 0.10], [0.05, 0.08, 0.10, 1.00],
])
factor_covariance = np.outer(factor_vol, factor_vol) * factor_correlation
specific_vol = np.where(universe["Sector"].eq("Corporate"), 0.010, 0.003)
cost_rate = universe["Trading cost (bp)"].to_numpy() / 10_000
benchmark_duration = benchmark_weights @ universe["Duration"]

def tracking_variance(weights):
    active = weights - benchmark_weights
    active_factors = active @ exposure_matrix
    return active_factors @ factor_covariance @ active_factors + np.sum((active * specific_v

def objective(weights):

```

```

turnover_penalty = 0.002 * np.sum((weights - current_weights) ** 2)
transaction_cost = 0.20 * np.sum(cost_rate * np.abs(weights - current_weights))
carry_reward = 0.0015 * weights @ universe["Yield"]
return tracking_variance(weights) + turnover_penalty + transaction_cost - carry_reward

treasury_mask = universe["Sector"].eq("Treasury").to_numpy()
corporate_mask = universe["Sector"].eq("Corporate").to_numpy()
constraints = [
    {"type": "eq", "fun": lambda w: w.sum() - 1},
    {"type": "ineq", "fun": lambda w: w[treasury_mask].sum() - treasury_minimum},
    {"type": "ineq", "fun": lambda w: corporate_cap - w[corporate_mask].sum()},
    {"type": "ineq", "fun": lambda w: duration_tolerance -
     abs(w @ universe["Duration"].to_numpy() - benchmark_duration)},
]
solution = minimize(objective, current_weights, method="SLSQP",
                    bounds=[(0, position_cap)] * len(universe), constraints=constraints,
                    options={"ftol": 1e-12, "maxiter": 2_000})
assert solution.success, solution.message
optimized_weights = solution.x

```

6 6. Compare the Alternatives

```

portfolios = {"Benchmark": benchmark_weights, "Current": current_weights,
              "Cell sampled": sampled_weights, "Optimized": optimized_weights}
comparison = pd.DataFrame({name: portfolio_summary(weights)
                           for name, weights in portfolios.items()}).T
comparison["Ex ante tracking error (%)"] = [
    100 * np.sqrt(tracking_variance(weights)) for weights in portfolios.values()]
comparison["Turnover (%)"] = [
    100 * np.abs(weights - current_weights).sum() / 2 for weights in portfolios.values()]
comparison["Estimated trading cost ($)"] = [
    portfolio_value * np.sum(cost_rate * np.abs(weights - current_weights))
    for weights in portfolios.values()]
comparison.round(3)

```

	Yield (%)	Duration	Convexity	Spread duration	Treasury weight (%)	Corporate weight
Benchmark	4.944	4.842	31.455	3.166	33.000	48.0

	Yield (%)	Duration	Convexity	Spread duration	Treasury weight (%)	Corporate weight
Current	4.962	4.610	28.776	3.187	30.000	52.0
Cell sampled	4.831	4.495	27.932	2.242	52.000	48.0
Optimized	4.915	4.792	31.024	3.068	34.805	45.0

```

active_exposures = pd.DataFrame({
    name: (weights - benchmark_weights) @ exposure_matrix
    for name, weights in portfolios.items() if name != "Benchmark"
}, index=factor_columns)
ax = active_exposures.plot(kind="bar", figsize=(8.2, 4.2))
ax.axhline(0, color="black", linewidth=0.8)
ax.set(ylabel="Active exposure (years)", title="Active factor exposures versus benchmark")
ax.grid(axis="y", alpha=0.3)
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

```

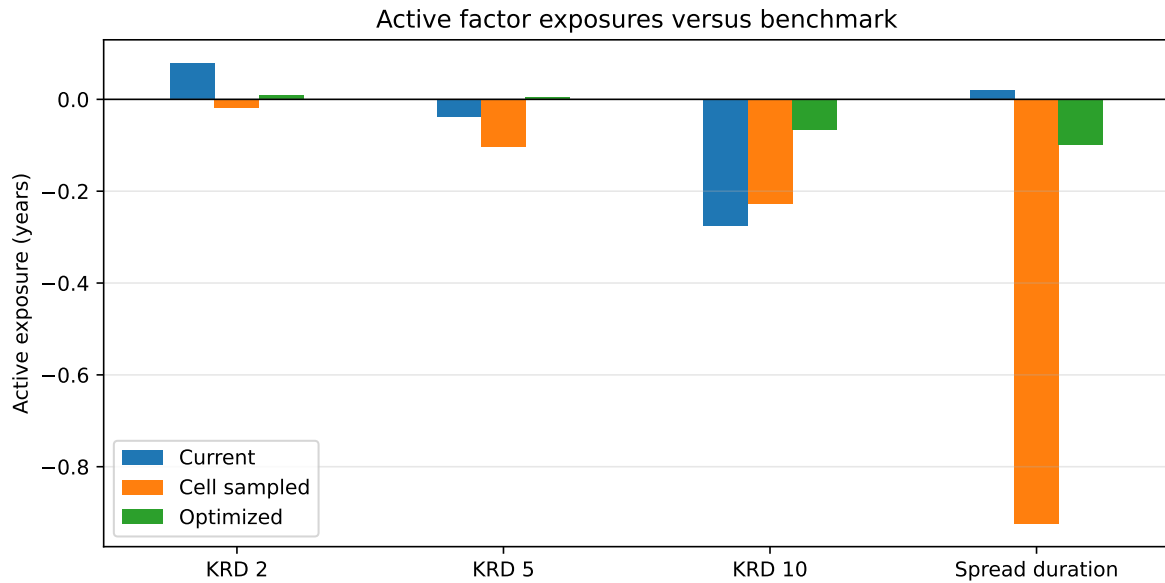


Figure 1: Optimization controls factor mismatches that cell weights alone can conceal

7 7. Turn Weights into Trades

Trades below are dollar market values. A real order list would convert market value to face amount using clean price, accrued interest, minimum denominations, round lots, and available dealer inventory.

```
trade_list = universe[["Bond", "Sector", "Cell"]].copy()
trade_list["Current weight (%)"] = 100 * current_weights
trade_list["Target weight (%)"] = 100 * optimized_weights
trade_list["Trade ($mm)"] = portfolio_value * (optimized_weights - current_weights) / 1_000_000
trade_list["Estimated cost ($)"] = portfolio_value * np.abs(optimized_weights - current_weights)
trade_list.loc[trade_list["Trade ($mm)"].abs().gt(0.025)].round(3)
```

	Bond	Sector	Cell	Current weight (%)	Target weight (%)	Trade (\$mm)	Estimated cost (\$)
0	UST 2Y	Treasury	Gov 1-4	12.0	12.416	0.208	20.778
1	UST 5Y	Treasury	Gov 5-8	10.0	11.360	0.680	136.01
2	UST 10Y	Treasury	Gov 9+	8.0	11.029	1.515	454.36
3	Agency 3Y	Agency	Gov 1-4	10.0	10.380	0.190	57.047
4	Agency 7Y	Agency	Gov 5-8	8.0	9.815	0.907	453.70
5	Bank A 3Y	Corporate	Corp 1-4	8.0	4.341	-1.830	1097.8
6	Bank B 5Y	Corporate	Corp 5-8	10.0	8.128	-0.936	655.03
9	Industrial B 5Y	Corporate	Corp 5-8	8.0	6.531	-0.735	587.62

8 8. Stress-Test the Decision

For Treasury yield shift Δy_i and spread change Δs_i ,

$$\frac{\Delta P_i}{P_i} \approx -D_i \Delta y_i - SD_i \Delta s_i + \frac{1}{2} C_i (\Delta y_i)^2.$$

```
scenarios = {
    "Parallel rates +50 bp": {"short": 50, "intermediate": 50, "long": 50, "spread": 0},
    "Bear steepener": {"short": 20, "intermediate": 45, "long": 80, "spread": 15},
    "Bull flattener": {"short": -20, "intermediate": -55, "long": -90, "spread": -10},
    "Credit selloff": {"short": 0, "intermediate": 0, "long": 0, "spread": 100},
}

def scenario_return(weights, scenario):
    rate_shock = np.where(universe["Duration"].lt(3.5), scenario["short"],
```

```

        np.where(universe["Duration"].lt(6.5), scenario["intermediate"], scenario["intermediate"])
    spread_shock = np.where(universe["Sector"].eq("Corporate"), scenario["spread"],
        np.where(universe["Sector"].eq("Agency"), scenario["spread"] * 0.5, scenario["spread"]))
    security_return = (-universe["Duration"].to_numpy() * rate_shock
        - universe["Spread duration"].to_numpy() * spread_shock
        + 0.5 * universe["Convexity"].to_numpy() * rate_shock**2)
    return weights @ security_return

scenario_results = pd.DataFrame({
    name: {scenario_name: portfolio_value * scenario_return(weights, shock) / 1_000_000
        for scenario_name, shock in scenarios.items()}
    for name, weights in portfolios.items()})
scenario_results.index.name = "Scenario P&L ($mm)"
scenario_results.round(3)

```

	Benchmark	Current	Cell sampled	Optimized
Scenario P&L (\$mm)				
Parallel rates +50 bp	-1.191	-1.134	-1.106	-1.179
Bear steepener	-1.463	-1.345	-1.374	-1.419
Bull flattener	1.682	1.535	1.584	1.635
Credit selloff	-1.284	-1.326	-1.121	-1.225

8.1 Explore rate and spread shocks (HTML)

The table above is the static PDF fallback. In HTML, change the parallel Treasury-rate and corporate-spread shocks to stress the optimized portfolio.

9 9. Validation and Recommendation

```

optimized_summary = portfolio_summary(optimized_weights)
assert np.isclose(optimized_weights.sum(), 1, atol=1e-8)
assert optimized_weights.min() >= -1e-9
assert optimized_weights.max() <= position_cap + 1e-8
assert optimized_weights[treasury_mask].sum() >= treasury_minimum - 1e-8
assert optimized_weights[corporate_mask].sum() <= corporate_cap + 1e-8
assert abs(optimized_summary["Duration"] - benchmark_duration) <= duration_tolerance + 1e-8
assert np.isfinite(comparison.to_numpy()).all()

```

```
print(f"Optimization converged: {solution.success}")
pd.Series({
    "Weights sum to": optimized_weights.sum(),
    "Duration difference vs benchmark": optimized_summary["Duration"] - benchmark_duration,
    "Treasury minimum slack": optimized_weights[treasury_mask].sum() - treasury_minimum,
    "Corporate cap slack": corporate_cap - optimized_weights[corporate_mask].sum(),
    "Position-cap slack": position_cap - optimized_weights.max(),
}).round(6)
```

Optimization converged: True

Weights sum to	1.0000
Duration difference vs benchmark	-0.0500
Treasury minimum slack	0.0980
Corporate cap slack	0.0000
Position-cap slack	0.0558
dtype:	float64

The optimizer is a disciplined starting point, not an automatic trade instruction. The manager should prefer it to the cell sample only if its incremental carry and factor control survive executable-price checks, lot rounding, issuer review, and scenario analysis. Rebalance when the benefit of restoring the mandate exceeds transaction costs and operational complexity.

10 Instructor Extension

Replace the embedded universe with a permitted bond inventory or evaluated-price snapshot. Add clean price and accrued interest, convert target market values to face amounts, impose issuer-level limits, and compare predicted tracking error with realized active returns over the next month.